

SYSTEM FOUNDATIONS

ShadowTalk AI Architecture Blueprint

Complete engineering specification: Component hierarchy, lifecycle state management, dual-backend hybrid model, and hardware performance profiling.

FOUNDER & ARCHITECT

Zain Ahmed Fahad Patel

ENGINE VERSION

v3.0.0 (Agentic Workspace)

FRAMEWORK

React 18 + Vite 5.4 + SWC

TYPE SYSTEM

TypeScript 5.8 Strict (0 Errors)

1 Philosophy, Vision & Evolution

Modern knowledge workers and developers suffer from severe cognitive fragmentation across disparate AI interfaces. Writing prompts in one tab, debugging code in an isolated playground, researching in a search engine, and managing documents in an external cloud produces friction and context loss. **ShadowTalk AI** solves this by synthesizing conversational intelligence, autonomous agents, in-browser code execution, and persistent memory into a single workspace.

DreamState

BACKGROUND AGENT

Executes long-horizon multi-turn tasks while the user is disconnected.

Omniscience

CONTEXT MEMORY

Connects facts across sessions and attached documents via adaptive vector graphs.

Shadow Twin

DIGITAL PERSONA

Reflects user operational style, tone, and logic with public shareable links.

The v3.0.0 Architecture Consolidation

In version 3.0.0, the repository underwent a major architectural pivot. Over 40 disparate legacy standalone pages were consolidated directly into the unified `/chatbot` canvas and modal command decks. This drastically reduced the routing footprint, accelerated code-splitting, and focused the application on agentic autonomy.

2 Application Lifecycle & Component Hierarchy

The application entry point is `src/main.tsx`, which boots critical polyfills, error interceptors, and device performance profiling before mounting the React tree:

Root Boot Sequence

- UUID Polyfill:** Ensures `crypto.randomUUID` is defined even in unauthenticated LAN/IP testing contexts.
- Global Error Interception:** Registers `setupGlobalErrorHandling()` to catch unhandled promise rejections and operational API errors.
- Hardware Tier Detection:** Invokes `applyPerfProfile()` synchronously before CSS paints to calibrate performance tier attributes.
- Chunk Recovery:** Installs `installViteChunkRecovery()` to automatically refresh stale dynamic import hashes after production deployments.
- Idle Warmup:** Defers non-critical WebGPU probing and service worker registration using `window.requestIdleCallback`.

Global Provider Hierarchy

The provider tree wrapped in `src/App.tsx` guarantees end-to-end security, state coordination, and theming:

```
<ErrorBoundary>
  <QueryClientProvider client={queryClient}>
    <ThemeProvider attribute="class" defaultTheme="dark" forcedTheme="dark">
      <TooltipProvider>
        <AuthProvider>
          <StealthKillSwitchProvider>
            <SecurityProvider>
              <ShadowMemoryProvider>
                <AutoImproveProvider>
```

```
<ThemeTemplateProvider>
  <CommandPaletteContext.Provider>
    <BrowserRouter>
      <SiteMotionProvider>
        <SitePageShell>
          <AnimatedRoutes />
        </SitePageShell>
      </SiteMotionProvider>
    </BrowserRouter>
  </CommandPaletteContext.Provider>
</ThemeTemplateProvider>
</AutoImproveProvider>
</ShadowMemoryProvider>
</SecurityProvider>
</StealthKillSwitchProvider>
</AuthProvider>
</TooltipProvider>
</ThemeProvider>
</QueryClientProvider>
</ErrorBoundary>
```

3 Dual-Backend Hybrid Storage Architecture

ShadowTalk AI employs a dual-tier storage strategy designed to function seamlessly in both 100% offline air-gapped environments and fully connected cloud environments:

Storage Tier	Underlying Technology	Scope & Responsibility
Local-First Layer	IndexedDB (<code>idb</code>), LocalStorage, Web Crypto API	Encrypted personal vault, adaptive turn memories, BYOK API keys, and guest conversation caches. Operates with zero network dependency.
Cloud Persistence	Firebase Firestore, Firebase Auth, Cloud Functions v2, Storage	Multi-device sync, verified user roles, audit trails, and shared LLM proxy quotas.

The Supabase PostgREST Adapter Layer

To avoid rewriting database queries across hundreds of components during backend migrations, ShadowTalk implements a transparent adapter in `src/integrations/firebase/adapter.ts` and `firestore.ts`:

Adapter Mechanics

When components invoke `backend.from('conversations').select('*').eq('user_id', uid).order('created_at', { ascending: false }).limit(20)`:

- The query builder translates SQL-like clauses into Firestore `query()`, `where()`, `orderBy()`, and `limit()` constraints.
- For operators unsupported natively in Firestore (e.g. `ilike`, arbitrary nested `is` nulls, complex OR groups), the adapter fetches matching document candidates and applies an in-memory post-filter via `matchesLocal()`.
- Timestamps formatted as Firestore `Timestamp` objects are automatically normalized into ISO-8601 strings, maintaining signature parity.

4 Hardware Performance & Low-End Degradation

Ensuring smooth 60 FPS operation across mobile devices, low-cost Chromebooks, and high-end workstations is handled by `src/lib/perf/devicePerfTier.ts`:

Tier Detection Factors

- VRAM & Max Buffer Size:** Probed via WebGPU adapter limits.
- Logical Cores:** `navigator.hardwareConcurrency`.
- Device Memory:** `navigator.deviceMemory` (GB).
- Power Preference:** Battery status API integration.

Automated Degraded Modes

- Tier 0 (Low-End):** Disables Three.js 3D canvas, turns off backdrop-filter blurs, simplifies spring transitions to standard CSS fades.
- Tier 1 (Standard):** Enables subtle motion, standard shadows.
- Tier 2 (High-End):** Full WebGL/Three.js pulse orbs, ambient canvas glow, real-time particle rendering.

